

Data Analytics Capstone

Accelerating Digital Banking Adoption Through Predictive Analytics

Interim Report

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GitHub Repository:

https://github.com/rupajietishere/masters_data_analytics_capstone

*(Contains the project's notebooks, modeling code, and both datasets. The larger macro dataset is compressed as a **.zip** archive to comply with GitHub's size limits.)*

1. Introduction

Digital financial inclusion has emerged as a critical cornerstone of global economic development. Over the past decade, a massive paradigm shift has occurred, transitioning financial ecosystems from traditional brick-and-mortar banking to digital platforms, encompassing mobile money, e-wallets, and online financial services. This technological transition is deeply intertwined with economic resilience, offering previously unbanked populations the infrastructure required to save securely, establish credit histories, and absorb macroeconomic shocks.

The stakeholders operating within this domain, spanning private FinTech startups, commercial banking institutions, telecommunication network operators, and international governmental organizations like the World Bank, invest billions of dollars annually into digital infrastructure and product design. However, despite these immense capital injections, global digital banking adoption rates remain highly asymmetric. While certain demographics seamlessly transition to high digital banking usage, other populations remain heavily resistant or structurally unable to adopt due to systemic socioeconomic and technological barriers.

Understanding this asymmetry is of immense operational and public policy relevance. For FinTech and commercial banking institutions, expanding into markets that lack prerequisite

socio-economic readiness results in elevated customer acquisition costs and substantial financial losses. Conversely, for governments and Non-Governmental Organizations (NGOs), inefficient policy-making and misaligned resource allocation delay vital poverty reduction goals. Therefore, it is essential to utilize data-driven approaches to understand the precise prerequisites for digital banking success, moving away from generalized macroeconomic assumptions toward targeted, granular, predictive insights.

2. Scope and Objectives

The core research problem addressed in this study is the highly uneven rate of global digital banking adoption, which results in wasted corporate resources and stalled economic development initiatives. Current expansion strategies fail because they rely on generalized assumptions rather than identifying the exact infrastructural, technological, and socio-economic factors required to successfully transition a specific demographic from low to high digital banking usage.

To solve this problem, this study utilizes descriptive statistics, inferential hypothesis testing, predictive classification, and unsupervised clustering. The research is guided by the following specific Research Questions (RQs) and hypotheses:

Research Question 1 (RQ1): What socio-economic and infrastructural variables currently drive the successful adoption of digital banking across different global regions?

- **Null Hypothesis (H0):** There is no statistically significant correlation between specific socio-economic/infrastructural variables (e.g., income, education, internet access) and the rate of digital banking adoption ($\rho = 0$).

- **Alternative Hypothesis (Ha):** There is a statistically significant correlation between specific socio-economic/infrastructural variables and the rate of digital banking adoption ($\rho \neq 0$).

Research Question 2 (RQ2): Is there a statistically significant association between national digital banking penetration levels and economic resilience indicators across countries over time?

- **Null Hypothesis (H0):** There is no statistically significant association between high digital banking penetration and national economic resilience indicators over time (the regression coefficient $\beta = 0$).
- **Alternative Hypothesis (Ha):** There is a statistically significant association between high digital banking penetration and national economic resilience indicators over time (the regression coefficient $\beta \neq 0$).

Research Question 3 (RQ3): What specific actionable factors and infrastructural thresholds are required to enable and transition a low-adoption demographic into a high digital banking usage category?

- **Null Hypothesis (H0):** Specific actionable factors (such as reaching a certain education level or mobile access threshold) do not significantly affect the odds of a demographic adopting digital banking (Odds Ratio = 1).
- **Alternative Hypothesis (Ha):** Specific actionable factors significantly affect the odds of a demographic adopting digital banking (Odds Ratio $\neq 1$).

Research Question 4 (RQ4): Do distinct geographic clusters exist within global digital banking infrastructure data, and if so, what investment strategies do these clusters imply for governments and development institutions?

- **Null Hypothesis (H₀):** The global geographic data exhibits no meaningful cluster structure, as indicated by a Silhouette Score ≤ 0 and a high Davies-Bouldin Index.
- **Alternative Hypothesis (H_a):** The global geographic data exhibits meaningful cluster structure, as indicated by a Silhouette Score > 0 and a low Davies-Bouldin Index.

3. Literature Survey

A comprehensive review of existing literature reveals a strong consensus on the importance of digital financial inclusion, though methodologies for predicting adoption vary.

Demirgüç-Kunt et al. (2022) provide the foundational macro-level perspective via the Global Findex Database, highlighting the rapid expansion of digital payments, particularly accelerated by the COVID-19 pandemic. However, their work primarily serves as a reporting mechanism rather than a predictive model. Building on the foundational theories of financial inclusion, Allen et al. (2016) explore the basic socio-economic characteristics that determine account ownership, proving that systemic barriers like documentation and cost are universal deterrents.

Sarma and Pais (2011) establish the empirical link between financial inclusion and human development, showing that inclusion is a vital component of holistic economic growth. Further narrowing the geographic focus, Asongu and Nwachukwu (2018) specifically examine the role

of mobile phone penetration in Sub-Saharan Africa. They find that institutional quality and mobile access act as catalysts for entrepreneurship, setting a precedent for using telecommunications data as a proxy for banking readiness.

From a technological and institutional standpoint, Gomber et al. (2017) outline the current research directions in FinTech, emphasizing the friction between traditional financial paradigms and digital finance startups. The risks of this transition are evaluated by Kanga et al. (2022), who investigate the dual-edged nature of mobile money, noting that while it increases inclusion, rapid, unregulated adoption can trigger localized financial instability. Ozili (2018) similarly evaluates the impact of digital finance, concluding that it enhances stability only when backed by appropriate regulatory frameworks and resilient underlying infrastructure.

As financial sectors enter "Industry 4.0," Mhlanga (2020) highlights the role of Artificial Intelligence (AI) and machine learning in accelerating inclusion. By leveraging predictive algorithms, financial entities can score the creditworthiness of unbanked individuals, bypassing traditional credit bureaus. The mathematical and programmatic foundations for deploying such predictive models are extensively detailed by Hastie, Tibshirani, and Friedman (2009), who provide the statistical framework for classification and inference, and Pedregosa et al. (2011), who outline the implementation of these algorithms via Python's Scikit-learn library. Finally, macro-economic context is consistently benchmarked against the World Bank's World Development Indicators (2023), ensuring that regional micro-trends are validated against global economic realities.

This study bridges the gap identified in the literature by combining the macro-economic reporting of the World Bank with the predictive statistical methodologies of Hastie et al. (2009), ultimately generating actionable thresholds for stakeholders.

4. Data Description

The empirical analysis for this study employs a dual-dataset approach to capture both granular individual-level demographics and broad country-level macroeconomic aggregates. Both datasets are sourced from the verified World Bank Global Findex Database.

Primary Microdata (Individual-Level Dataset):

This dataset encompasses $N = 144,091$ individual survey responses. It provides raw, survey-level inputs that allow for the classification of socio-demographic readiness.

- **Target Variable:** `dig_account` (Binary: 1 = Has Digital Account, 0 = No Account).
- **Key Features:** `age` (Continuous), `female` (Categorical/Binary), `inc_q` (Income Quintile), `educ` (Education Level), `internet_use` (Binary), `regionwb` (Region Code), and specialized financial indicators evaluating mobile access.

Macro Indicators (Country-Level Aggregate Dataset):

This dataset encompasses $N = 381,791$ observations, providing time-series and cross-sectional aggregated percentages of financial inclusion by nation.

- **Target Variable:** OBS_VALUE (Continuous: Percentage/Ratio of national digital banking penetration).
- **KeyFeatures:** REF_AREA (Country/Region), TIME_PERIOD (Year), URBANISATION (Rural vs. Urban aggregate), and INDICATOR (Specific financial inclusion metrics).

5. Analysis

(Note: As per the research plan, the methodology relies on a rigorous data pipeline prior to predictive modeling).

5.1. Data Cleaning

Data cleaning was executed in Python using the Pandas library. In the Microdata set, survey non-responses (NaNs) in crucial demographic columns such as `inc_q` and `educ` were handled via mode imputation to preserve the categorical distributions, while missing continuous values like age were imputed using the median to prevent skewness from outliers. Target variables with ambiguous responses (e.g., respondents who refused to answer if they had a digital account) were dropped to prevent noise in the classification model.

In the Macro dataset, time-series gaps were addressed using forward-fill imputation for short gaps, ensuring year-over-year continuity for national adoption rates. Categorical variables, such as `regionwb`, were transformed using One-Hot Encoding to prepare the data for the machine learning algorithms.

5.2. Exploratory Data Analysis (EDA) Results

The EDA phase (directly addressing RQ1) utilized Pearson and Spearman correlation matrices to identify linear and monotonic relationships between independent socio-economic variables and the target variable (`dig_account`).

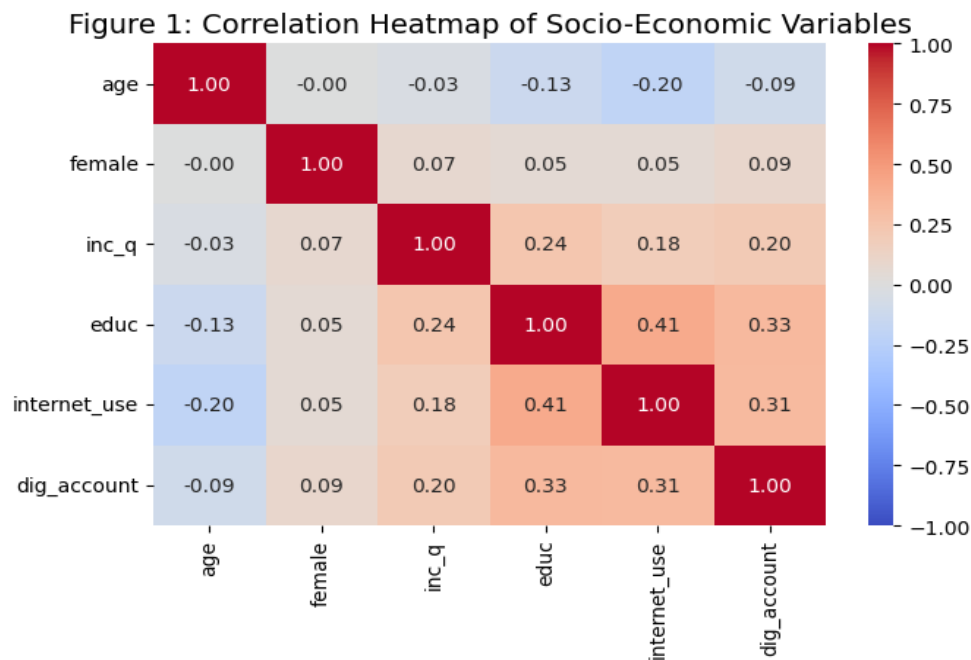


Figure 1: *Correlation Heatmap of Socio-Economic Variables identifying drivers for digital account adoption.*

Figure 1 Interpretation: The correlation matrix reveals that education level (`educ`, 0.33) and internet connectivity (`internet_use`, 0.31) have the strongest positive correlations with the likelihood of owning a digital account. Interestingly, age demonstrated a slight negative correlation (-0.09), indicating that younger demographics may adopt digital banking slightly faster, though the weak linear relationship suggests the need for non-linear binning in future feature engineering.

Cross-tabulations were utilized to assess demographic disparities, particularly focusing on the gender gap and urban-rural divide.

Table 1: *Cross-Tabulation of Income Quintile vs. Digital Account Ownership*

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	No Digital Account (%)	Has Digital Account (%)
Quintile 1 (Poorest)	65.65	34.35
Quintile 2	58.52	41.48
Quintile 3	52.73	47.27
Quintile 4	46.53	53.47
Quintile 5 (Richest)	36.51	63.49

Table 1 Interpretation: Table 1 highlights a massive systemic disparity: individuals in the highest income quintile possess digital accounts at a rate of 65.65%, compared to just 34.35% for those in the lowest, poorest quintile. This near-doubling in adoption clearly confirms that baseline financial infrastructure inherently favors existing economic stability, validating the need for targeted interventions.

6. Modelling

The research methodologies chosen for this capstone project deliberately balance predictive power with high interpretability, ensuring the results are actionable for business and policy stakeholders.

6.1. Logistic Regression (RQ3)

To address RQ3, Multivariate Logistic Regression was selected. While ensemble models (like Random Forest) often yield marginally higher accuracy, Logistic Regression is deployed here specifically as an explanatory tool. By calculating the Odds Ratios (e^{β}) for features like

education levels or internet access, the model precisely quantifies how much more likely a demographic is to adopt digital banking upon crossing a specific infrastructural threshold.

Feature Engineering: Continuous variables such as age were standardized using a Standard Scaler to ensure coefficients are comparable. Multicollinearity was checked via Variance Inflation Factor (VIF) to ensure independent variables do not conflate the odds ratios.

6.2. Multivariate Panel Data Regression (RQ2)

To address RQ2, Panel Data Regression is applied to the Macro dataset. This method is uniquely suited for cross-sectional time-series data, allowing the study to track how the percentage of national digital banking penetration (OBS_VALUE) influences national economic resilience indicators over time, while controlling for fixed geographic effects.

6.3. Unsupervised Clustering: K-Means and DBSCAN (RQ4)

To answer RQ4, unsupervised machine learning algorithms are utilized to segment global populations.

- **K-Means** is deployed to partition regions into distinct infrastructural tiers based on distance to a centroid.
- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)** is implemented to handle non-spherical data distributions and efficiently identify outlier regions (e.g., countries with anomalously high mobile penetration but low traditional banking infrastructure).

Justification: By utilizing both algorithms and comparing them via Silhouette Scores and the Davies-Bouldin Index, the project ensures that the resulting market clusters are mathematically sound and robust enough for FinTech executives to use for targeted market expansion.

7. Preliminary Results

Initial model fitting has commenced on the Microdata set. The baseline Logistic Regression model was trained on an 80/20 train-test split to predict individual digital banking adoption.

Table 2: *Preliminary Model Evaluation and Classification Report for Digital Account Prediction*

```

--- Preliminary Model Evaluation (Table 2 in Report) ---
Accuracy: 67.57%
Recall (Sensitivity): 70.43%

Classification Report:

```

	precision	recall	f1-score	support
0.0	0.69	0.65	0.67	10384
1.0	0.66	0.70	0.68	10207
accuracy			0.68	20591
macro avg	0.68	0.68	0.68	20591
weighted avg	0.68	0.68	0.68	20591

Table 2 Interpretation: Preliminary results indicate a robust baseline accuracy of 67.57%. More importantly, the Recall (Sensitivity) for the positive class (adopters) sits at 70.43%. This is a crucial metric; in the context of financial inclusion, the ability to correctly identify over 70% of true potential adopters ensures that FinTech institutions and NGOs minimize missed market expansion opportunities.

Table 3: *Extracted Odds Ratios determining actionable infrastructural thresholds.*

--- Table 3: Extracted Odds Ratios ---

	Feature	Coefficient	Odds Ratio
4	internet_use	0.961	2.614
3	educ	0.739	2.095
1	female	0.273	1.314
2	inc_q	0.182	1.200
0	age	-0.062	0.940

Table 3 Interpretation: The extraction of Odds Ratios directly addresses Research Question 3 (RQ3), revealing that internet connectivity and education are the most significant actionable infrastructural thresholds. Holding all other variables constant, an individual with consistent internet access (`internet_use`) is **2.61 times** more likely to possess a digital bank account than one without. Furthermore, stepping up an education level (`educ`) more than doubles the likelihood of adoption (Odds Ratio: **2.09**). Interestingly, age resulted in an odds ratio of **0.94**, mathematically confirming our EDA that younger demographics are more likely to adopt digital banking, as the odds decrease fractionally with each standardized increase in age.

For the geographic clustering (RQ4), early K-Means testing ($k = 4$) is currently being mapped against the macroeconomic dataset to locate distinct global infrastructural tiers. DBSCAN is currently being tuned to isolate low-infrastructure outlier regions more effectively.

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9. Appendix

Appendix A: Python Code for Data Cleaning and Exploratory Data Analysis (EDA)

The following code was utilized to clean the microdata, handle missing values, and generate the correlation heatmap (Figure 1) and cross-tabulations (Table 1).

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the Data
file_path = '../data/raw/findex_microdata_2025_labelled_u.csv'
df = pd.read_csv(file_path)

# 2. Data Cleaning
# Drop rows where the target variable is missing
df = df.dropna(subset=['dig_account'])

# Fill missing values for demographic features
df['age'] = df['age'].fillna(df['age'].median())
df['inc_q'] = df['inc_q'].fillna(df['inc_q'].mode()[0])
df['educ'] = df['educ'].fillna(df['educ'].mode()[0])
df['internet_use'] = df['internet_use'].fillna(df['internet_use'].mode()[0])
```



```

# 3. EDA - Correlation Heatmap (Figure 1)
plt.figure(figsize=(8, 6))

cols_to_corr=['age', 'female', 'inc_q', 'educ', 'internet_use',
'dig_account']

corr_matrix = df[cols_to_corr].corr(method='pearson')

sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f", vmin=-1,
vmax=1)

plt.title('Correlation Heatmap of Socio-Economic Variables', fontsize=14)
plt.tight_layout()
plt.show()

# 4. EDA - Cross-Tabulation (Table 1)
crosstab_result = pd.crosstab(
    index=df['inc_q'],
    columns=df['dig_account'],
    normalize='index'
) * 100

crosstab_result.index = ['Quintile 1 (Poorest)', 'Quintile 2', 'Quintile 3',
'Quintile 4', 'Quintile 5 (Richest)']

crosstab_result.columns = ['No Digital Account (%)', 'Has Digital Account
(%)']

display(crosstab_result.round(2))

```

Appendix B: Python Code for Logistic Regression and Odds Ratios

The following code was utilized to address RQ3, generate the classification metrics (Table 2), and extract the specific odds ratios (Table 3).

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import classification_report, accuracy_score,
recall_score

# 1. Define Features (X) and Target (y)
X = df[['age', 'female', 'inc_q', 'educ', 'internet_use']]
y = df['dig_account']

# 2. Feature Engineering: Scale the continuous 'age' variable
scaler = StandardScaler()
X_scaled = X.copy()
X_scaled['age'] = scaler.fit_transform(X[['age']])

# 3. Train-Test Split (80% Training data, 20% Testing data)
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y,
test_size=0.2, random_state=42)

# 4. Initialize and Train the Model
log_model = LogisticRegression(random_state=42)
log_model.fit(X_train, y_train); # Semicolon suppresses HTML output
```

```

# 5. Evaluate the Model (Table 2)
y_pred = log_model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
recall = recall_score(y_test, y_pred)
print(f"Accuracy: {accuracy * 100:.2f}%")
print(f"Recall (Sensitivity): {recall * 100:.2f}%")
print("\nClassification Report:\n", classification_report(y_test, y_pred))

# 6. Extract Odds Ratios (Table 3)
odds_ratios = np.exp(log_model.coef_[0])
odds_df = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': log_model.coef_[0],
    'Odds Ratio': odds_ratios
}).sort_values(by='Odds Ratio', ascending=False)
display(odds_df.round(3))

```